$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{0^+} & \mathcal{K}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{0^+}\dagger & 0 & 0 \\
\mathcal{K}^{\#1}_{0^-}\dagger^{\alpha\beta\chi} & 0 & 3k^2 \overset{(4)}{\kappa}_{15} + 3\overset{(2)}{\kappa}_{1}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{K}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{1^-}{}_{\alpha} & \mathcal{K}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{K}^{\#1}_{1^+}\dagger^{\alpha\beta} & \frac{1}{2}(-\mathrm{Y}_1 k^2 + 4\overset{(2)}{\kappa}_{1}) & \frac{\mathrm{Y}_2 k^2 + 4\overset{(2)}{\kappa}_{1}}{2\sqrt{2}} & 0 & 0 \\
\mathcal{K}^{\#2}_{1^+}\dagger^{\alpha\beta} & \frac{\mathrm{Y}_2 k^2 + 4\overset{(2)}{\kappa}_{1}}{2\sqrt{2}} & \frac{1}{2}(\mathrm{Y}_3 k^2 + 2\overset{(2)}{\kappa}_{1}) & 0 & 0 \\
\mathcal{K}^{\#1}_{1^-}\dagger^{\alpha} & 0 & 0 & -k^2 \overset{(4)}{\kappa}_{2} & -\sqrt{2}\, k^2 \overset{(4)}{\kappa}_{2} \\
\mathcal{K}^{\#2}_{1^-}\dagger^{\alpha} & 0 & 0 & -\sqrt{2}\, k^2 \overset{(4)}{\kappa}_{2} & -2k^2 \overset{(4)}{\kappa}_{2}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{K}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{K}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{K}^{\#1}_{2^+}\dagger^{\alpha\beta} & \frac{3\mathrm{Y}_4 k^2}{2} & 0 \\
\mathcal{K}^{\#1}_{2^-}\dagger^{\alpha\beta\chi} & 0 & 0
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{0^+} & \mathcal{J}^{\#1}_{0^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{0^+}\dagger & 0 & 0 \\
\mathcal{J}^{\#1}_{0^-}\dagger^{\alpha\beta\chi} & 0 & \frac{1}{3(k^2 \overset{(4)}{\kappa}_{15} + \overset{(2)}{\kappa}_{1})}
\end{array}
$$

$$
\begin{array}{c|cccc}
 & \mathcal{J}^{\#1}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#2}_{1^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{1^-}{}_{\alpha} & \mathcal{J}^{\#2}_{1^-}{}_{\alpha} \\
\hline
\mathcal{J}^{\#1}_{1^+}\dagger^{\alpha\beta} & \frac{\mathrm{Y}_3 k^2 + 2\overset{(2)}{\kappa}_{1}}{2\,\mathrm{Det}(1^+)} & \frac{-\mathrm{Y}_2 k^2 - 4\overset{(2)}{\kappa}_{1}}{2\sqrt{2}\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#2}_{1^+}\dagger^{\alpha\beta} & \frac{-\mathrm{Y}_2 k^2 - 4\overset{(2)}{\kappa}_{1}}{2\sqrt{2}\,\mathrm{Det}(1^+)} & \frac{-\mathrm{Y}_1 k^2 + 4\overset{(2)}{\kappa}_{1}}{2\,\mathrm{Det}(1^+)} & 0 & 0 \\
\mathcal{J}^{\#1}_{1^-}\dagger^{\alpha} & 0 & 0 & -\frac{1}{9k^2 \overset{(4)}{\kappa}_{2}} & -\frac{\sqrt{2}}{9k^2 \overset{(4)}{\kappa}_{2}} \\
\mathcal{J}^{\#2}_{1^-}\dagger^{\alpha} & 0 & 0 & -\frac{\sqrt{2}}{9k^2 \overset{(4)}{\kappa}_{2}} & -\frac{2}{9k^2 \overset{(4)}{\kappa}_{2}}
\end{array}
$$

$$
\begin{array}{c|cc}
 & \mathcal{J}^{\#1}_{2^+}{}_{\alpha\beta} & \mathcal{J}^{\#1}_{2^-}{}_{\alpha\beta\chi} \\
\hline
\mathcal{J}^{\#1}_{2^+}\dagger^{\alpha\beta} & \frac{1}{\mathrm{Det}(2^+)} & 0 \\
\mathcal{J}^{\#1}_{2^-}\dagger^{\alpha\beta\chi} & 0 & 0
\end{array}
$$

Abbreviations used in matrices

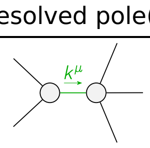
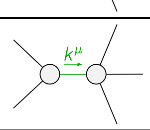
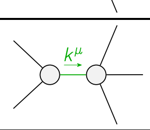
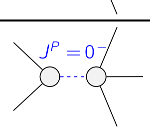
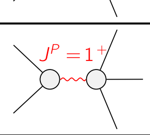
$$
\begin{aligned}
&\mathrm{Y}_1 == 3\overset{(4)}{\kappa}_{1} - 4\overset{(4)}{\kappa}_{15} + 3\overset{(4)}{\kappa}_{2} + 2\overset{(4)}{\kappa}_{3} \;\&\& \\
&\mathrm{Y}_2 == 4\overset{(4)}{\kappa}_{15} + \overset{(4)}{\kappa}_{4} \;\&\&\; \mathrm{Y}_3 == 2\overset{(4)}{\kappa}_{15} - 9\overset{(4)}{\kappa}_{2} + \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4} \;\&\&\; \mathrm{Y}_4 == \overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{2} \;\&\& \\
&\mathrm{Det}(1^+) == -\frac{1}{8} k^2 (k^2 (84\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{2} - 54\overset{(4)}{\kappa}_{2}{}^2 - 30\overset{(4)}{\kappa}_{2}\overset{(4)}{\kappa}_{3} + 4\overset{(4)}{\kappa}_{3}{}^2 + 6\overset{(4)}{\kappa}_{2}\overset{(4)}{\kappa}_{4} + 4\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2 + \\
&6\overset{(4)}{\kappa}_{1}(2\overset{(4)}{\kappa}_{15} - 9\overset{(4)}{\kappa}_{2} + \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})) + 12(\overset{(4)}{\kappa}_{1} + 7\overset{(4)}{\kappa}_{2})\overset{(2)}{\kappa}_{1}) \;\&\&\; \mathrm{Det}(2^+) == \frac{3}{2} k^2 (\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{2})
\end{aligned}
$$

Lagrangian

$$
\begin{aligned}
&\overset{(2)}{\kappa}_{1}\, \mathcal{K}_{\alpha\beta\chi}\, \mathcal{K}^{\alpha\beta\chi} - 2\overset{(2)}{\kappa}_{1}\, \mathcal{K}^{\alpha\beta\chi}\, \mathcal{K}_{\beta\alpha\chi} + \overset{(4)}{\kappa}_{1}\, \partial_\beta \mathcal{K}^{\delta}{}_{\chi\delta}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \\
&\overset{(4)}{\kappa}_{2}\, \partial_\chi \mathcal{K}^{\delta}{}_{\beta\delta}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta} + \overset{(4)}{\kappa}_{3}\, \partial_\beta \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\alpha\chi}{}^{\delta} + \overset{(4)}{\kappa}_{4}\, \partial_\alpha \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\beta\chi}{}^{\delta} - 3\overset{(4)}{\kappa}_{1}\, \partial_\beta \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} - \\
&3\overset{(4)}{\kappa}_{2}\, \partial_\beta \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} - \overset{(4)}{\kappa}_{3}\, \partial_\beta \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}_{\chi\alpha}{}^{\delta} - 6\overset{(4)}{\kappa}_{2}\, \partial^\chi \mathcal{K}^{\alpha}{}_{\alpha}{}^{\beta}\, \partial_\delta \mathcal{K}_{\chi\beta}{}^{\delta} - \frac{9}{2}\overset{(4)}{\kappa}_{2}\, \partial_\alpha \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \\
&\frac{1}{2}\overset{(4)}{\kappa}_{3}\, \partial_\alpha \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \frac{1}{2}\overset{(4)}{\kappa}_{4}\, \partial_\alpha \mathcal{K}^{\alpha\beta\chi}\, \partial_\delta \mathcal{K}^{\delta}{}_{\beta\chi} + \overset{(4)}{\kappa}_{15}\, \partial_\delta \mathcal{K}_{\alpha\beta\chi}\, \partial^\delta \mathcal{K}^{\alpha\beta\chi} - 2\overset{(4)}{\kappa}_{15}\, \partial_\delta \mathcal{K}_{\beta\alpha\chi}\, \partial^\delta \mathcal{K}^{\alpha\beta\chi}
\end{aligned}
$$

Added source term(s):	$\mathcal{K}^{\alpha\beta\chi}\ \mathcal{J}_{\alpha\beta\chi}$

Source constraint(s)	# constraint(s)	Covariant form
$\mathcal{J}^{\#1}_{0^+} == 0$	1	$\partial_\beta \mathcal{J}^{\alpha}{}_{\alpha}{}^{\beta} == 0$
$\mathcal{J}^{\#1\alpha}_{1^-} == \mathcal{J}^{\#2\alpha}_{1^-}$	3	$\partial_\chi \partial^\alpha \mathcal{J}^{\beta}{}_{\beta}{}^{\chi} + \partial_\chi \partial^\chi \mathcal{J}^{\beta\alpha}{}_{\beta} == 0$
$\mathcal{J}^{\#1\alpha\beta\chi}_{2^-} == 0$	5	$3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\alpha \mathcal{J}^{\delta\beta\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\alpha \mathcal{J}^{\delta\beta}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\alpha\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\chi\alpha\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\beta \mathcal{J}^{\delta\alpha\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\beta\alpha\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\delta\alpha\beta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\alpha\beta\chi} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\alpha \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta \mathcal{J}^{\delta\beta\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon \mathcal{J}^{\delta\alpha}{}_{\delta} == 3\,\partial_\epsilon\partial_\delta\partial^\chi\partial^\beta \mathcal{J}^{\delta\alpha\epsilon} + 3\,\partial_\epsilon\partial^\epsilon\partial^\chi\partial^\beta \mathcal{J}^{\delta\alpha}{}_{\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\beta\chi\delta} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\chi\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\alpha \mathcal{J}^{\delta\beta\chi} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\chi \mathcal{J}^{\alpha\beta\delta} + 2\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\beta\alpha\chi} + 4\,\partial_\epsilon\partial^\epsilon\partial_\delta\partial^\delta \mathcal{J}^{\chi\alpha\beta} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\beta \mathcal{J}^{\delta}{}_{\delta}{}^{\epsilon} + 3\,\eta^{\beta\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial_\delta \mathcal{J}^{\delta\alpha\epsilon} + 3\,\eta^{\alpha\chi}\,\partial_\phi\partial^\phi\partial_\epsilon\partial^\epsilon \mathcal{J}^{\delta\beta}{}_{\delta}$
Total # constraint(s):	9	

Resolved pole(s)	# polarization(s)	Square mass	Residue
	2	0	$\frac{\overset{(4)}{\kappa}_{1}{}^2 + 6\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{2} + 12\overset{(4)}{\kappa}_{2}{}^2}{\overset{(4)}{\kappa}_{2}(\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{2})(\overset{(4)}{\kappa}_{1} + 7\overset{(4)}{\kappa}_{2})}$
	1	0	$-\frac{4\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{2} + 28\overset{(4)}{\kappa}_{2}{}^2 + \sqrt{(\overset{(4)}{\kappa}_{1} + 7\overset{(4)}{\kappa}_{2})^2(\overset{(4)}{\kappa}_{1}{}^2 + 6\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{2} + 33\overset{(4)}{\kappa}_{2}{}^2)}}{\overset{(4)}{\kappa}_{2}(\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{2})(\overset{(4)}{\kappa}_{1} + 7\overset{(4)}{\kappa}_{2})}$
	1	0	$\frac{-4\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{2} - 28\overset{(4)}{\kappa}_{2}{}^2 + \sqrt{(\overset{(4)}{\kappa}_{1} + 7\overset{(4)}{\kappa}_{2})^2(\overset{(4)}{\kappa}_{1}{}^2 + 6\overset{(4)}{\kappa}_{1}\overset{(4)}{\kappa}_{2} + 33\overset{(4)}{\kappa}_{2}{}^2)}}{\overset{(4)}{\kappa}_{2}(\overset{(4)}{\kappa}_{1} + \overset{(4)}{\kappa}_{2})(\overset{(4)}{\kappa}_{1} + 7\overset{(4)}{\kappa}_{2})}$
	1	$-\frac{\overset{(2)}{\kappa}_{1}}{\overset{(4)}{\kappa}_{15}}$	$-\frac{1}{3\overset{(4)}{\kappa}_{15}}$
	3	$-\frac{12(\overset{(4)}{\kappa}_{1} + 7\overset{(4)}{\kappa}_{2})\overset{(2)}{\kappa}_{1}}{84\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{2} - 54\overset{(4)}{\kappa}_{2}{}^2 - 30\overset{(4)}{\kappa}_{2}\overset{(4)}{\kappa}_{3} + 4\overset{(4)}{\kappa}_{3}{}^2 + 6\overset{(4)}{\kappa}_{2}\overset{(4)}{\kappa}_{4} + 4\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2 + 6\overset{(4)}{\kappa}_{1}(2\overset{(4)}{\kappa}_{15} - 9\overset{(4)}{\kappa}_{2} + \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})}$	$\frac{2(6\overset{(4)}{\kappa}_{1}{}^2 + 114\overset{(4)}{\kappa}_{2}{}^2 - 8\overset{(4)}{\kappa}_{2}(2\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4}) + (2\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4})^2 + 4\overset{(4)}{\kappa}_{1}(3\overset{(4)}{\kappa}_{2} + 2\overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4}))}{(\overset{(4)}{\kappa}_{1} + 7\overset{(4)}{\kappa}_{2})(84\overset{(4)}{\kappa}_{15}\overset{(4)}{\kappa}_{2} - 54\overset{(4)}{\kappa}_{2}{}^2 - 30\overset{(4)}{\kappa}_{2}\overset{(4)}{\kappa}_{3} + 4\overset{(4)}{\kappa}_{3}{}^2 + 6\overset{(4)}{\kappa}_{2}\overset{(4)}{\kappa}_{4} + 4\overset{(4)}{\kappa}_{3}\overset{(4)}{\kappa}_{4} + \overset{(4)}{\kappa}_{4}{}^2 + 6\overset{(4)}{\kappa}_{1}(2\overset{(4)}{\kappa}_{15} - 9\overset{(4)}{\kappa}_{2} + \overset{(4)}{\kappa}_{3} + \overset{(4)}{\kappa}_{4}))}$
Resolved unitarity condition(s):	(Demonstrably impossible)		